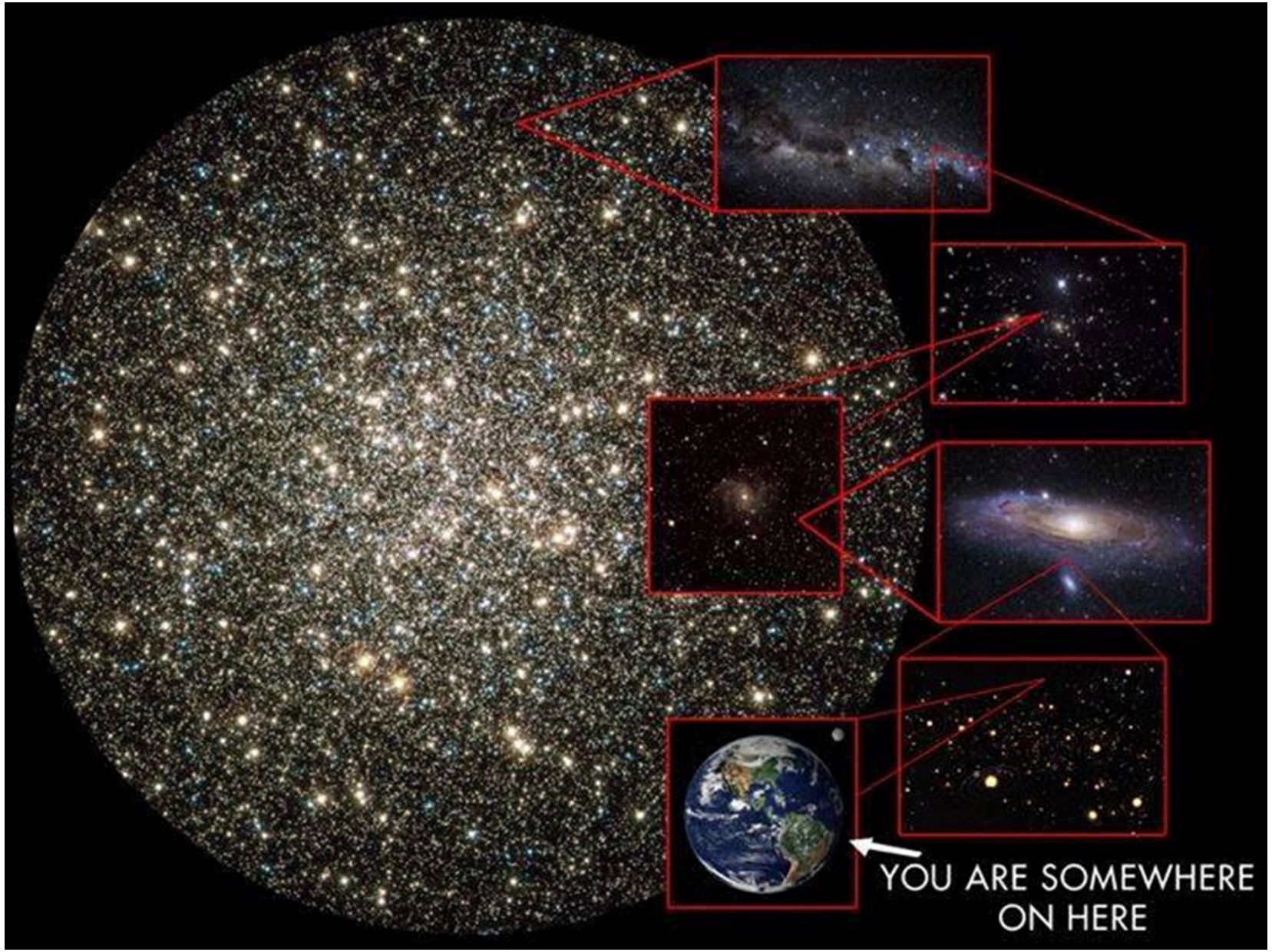
Introduction to Astronomy

What is astronomy?

- Astronomy is the scientific study of **celestial** objects (planets, moons, stars, galaxies, etc.)
- It is a way to understand the physical laws and origins of the **universe**
- The **universe** contains everything that exists – all matter, energy, and **space**

Where are we?



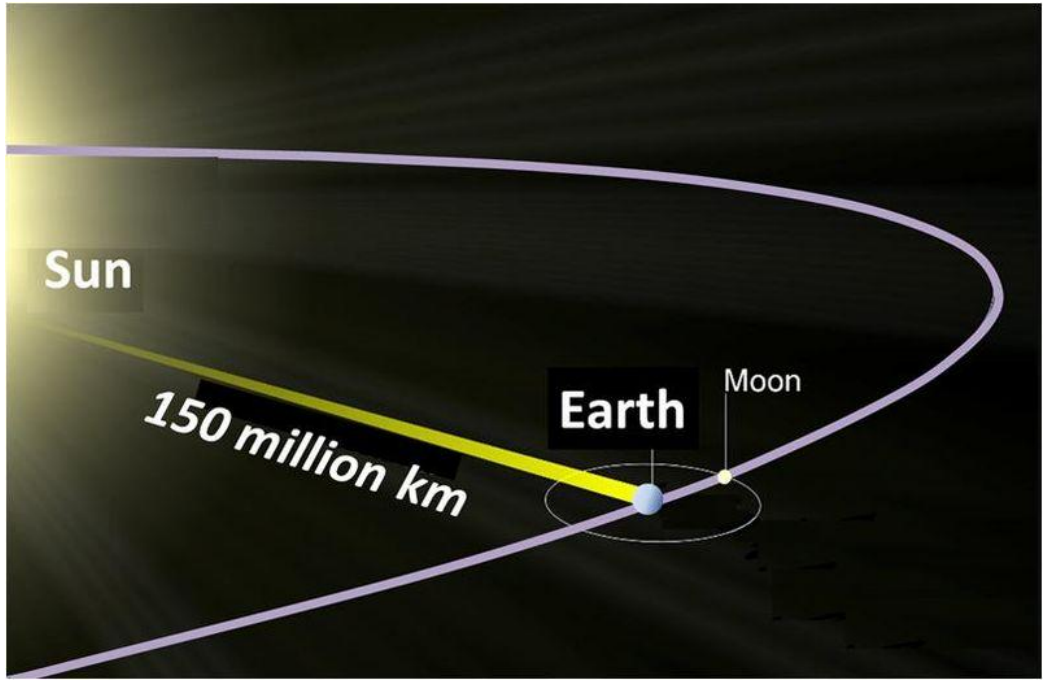
The Universe is Big

- Stars other than our Sun are so far distant that astronomers speak of their distances not in terms of kilometers – but in **light-years**
- A light-year is the distance light travels in one year - this is equal to **9.5** trillion km!
- It's just over **2** million light years to our nearest galactic neighbor, the Andromeda galaxy

One light-year
5.8 trillion miles
(9.4 trillion kilometers)

Distances in the Solar System

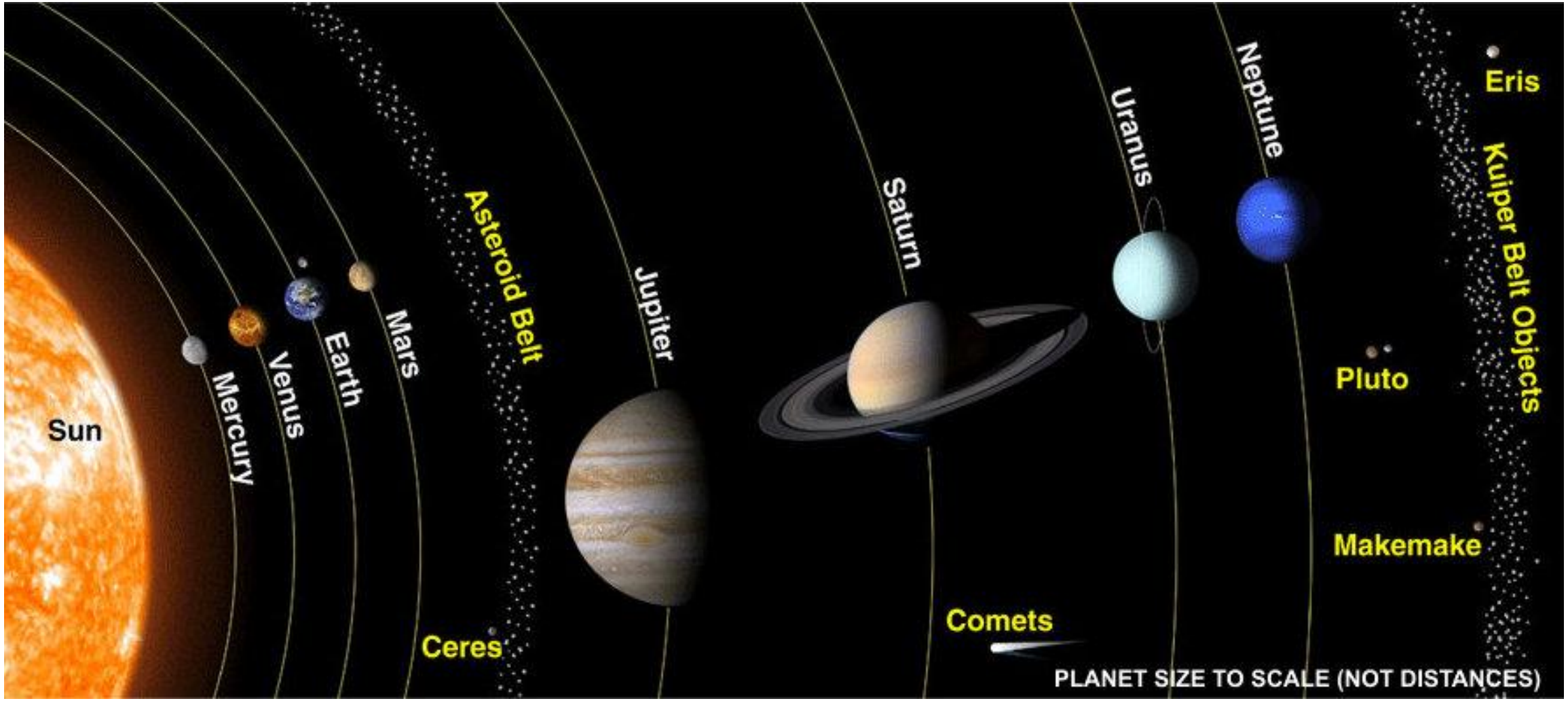
- Astronomers use **astronomical units** (AU)
- 1 AU equals about **150** million km
- Light from the sun takes **eight** minutes to get to us



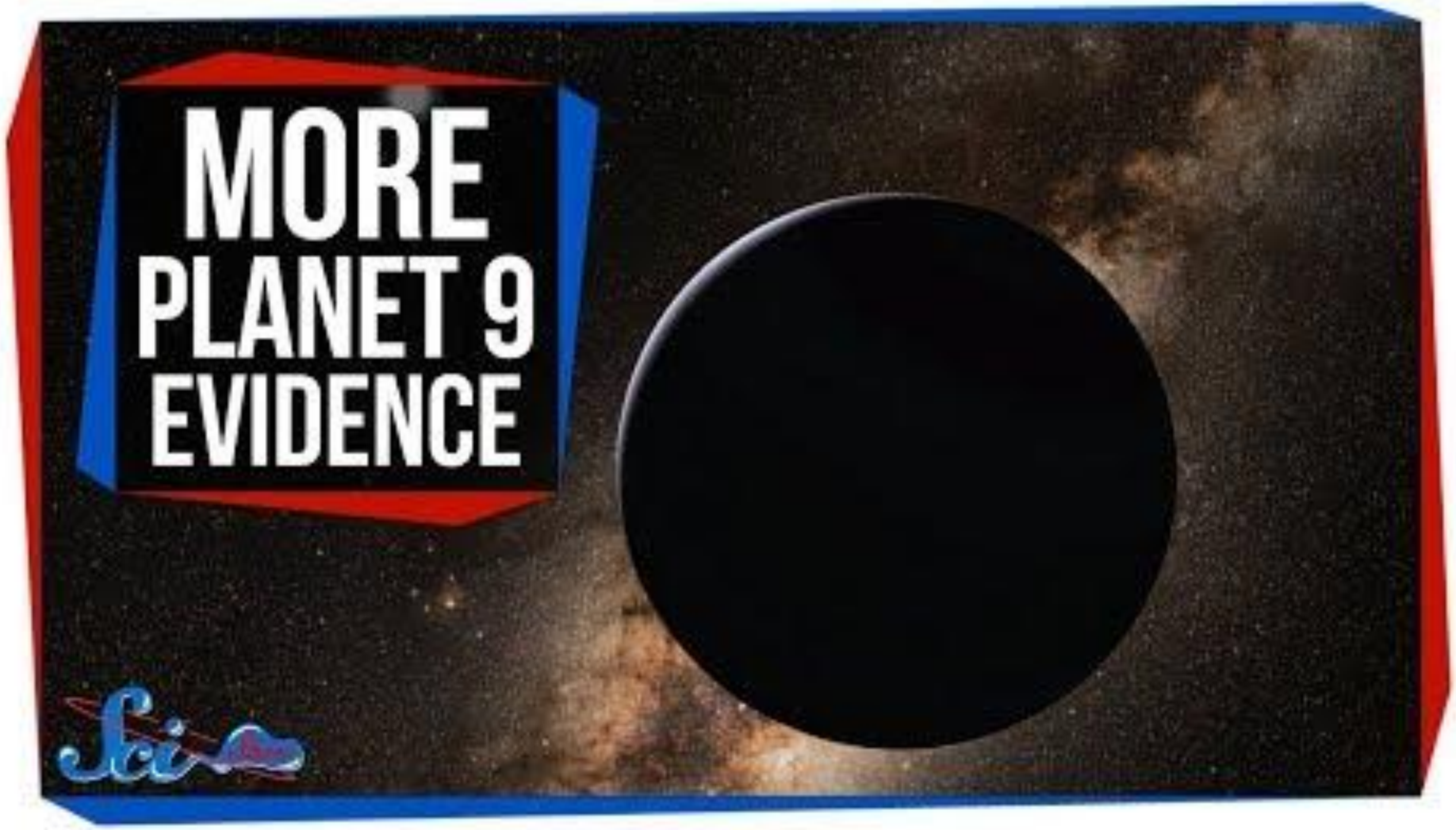
BEST OF THE MONTH
WINNER
NATIONAL GEOGRAPHIC
SHORT FILM SHOWCASE
OFFICIAL SELECTION
WINNER
To Scale:
THE SOLAR SYSTEM
OFFICIAL SELECTION
MOUNTAINFILM
2016
Official Selection
Anchorage International Film Festival
2016
OFFICIAL SELECTION
Durango Independent Film Festival

The Solar System

- Our solar system consists of our star, the Sun, and everything bound to it by **gravity**
- This includes, the planets, **dwarf** planets, moons, comets, and asteroids
- It also includes the **Kuiper Belt** that lies past Neptune's orbit
- This is a sparsely occupied ring of **icy bodies** (all smaller than Pluto)


Eris
Kuiper Belt Objects
Neptune
Uranus
Pluto
Makemake
Saturn
Comets
Jupiter
Asteroid Belt
Ceres
Mars
Earth
Venus
Mercury
Sun
PLANET SIZE TO SCALE (NOT DISTANCES)

Credit: NASA/Johns Hopkins University Applied Physics Laboratory/Southwest Research Institute


MORE
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EVIDENCE

Your turn

- Astronomy 101